

HARDWARE FOR AI ECE-510 CMAN 8

Given Data - $N = 1024$ output Neurons ($\log_2(1024) = 10 \text{ bits}$) $f = 50 \text{ Hz}$.

TASK 1

Formula $\Rightarrow R = N \times f$

$$= 1024 \times 50 \text{ Hz}$$

$$R = 51,200 \text{ spikes/second}$$

$$(51.2 \text{ kspikes/second})$$

TASK 2

Formula $\Rightarrow B = R \times 20 \text{ bits}$

$$= 51,200 \times 20$$

$$B = 1,024,000 \text{ bits/s}$$

$$(1.024 \text{ Mbit/s})$$

TASK 3

$$B = 1.024 \text{ Mbit/s}$$

Considering the bandwidth all the interfaces would be able to sustain the mean rate.

But considering the other tradeoffs I²C interface can be the ideal choice with simple interface with minimum pin count and design overhead.

TASK 4

Neurons firing in burst $= 1024 \times 0.25 = 256 \text{ neurons}$

Total bits generated $= 256 \text{ packets} \times 20 \text{ bits/packet} = 5120 \text{ bits}$

$$\text{Peak } B = \frac{5120}{0.001 \text{ s}} = 5,120,000 \text{ bits/s} \Rightarrow 5.12 \text{ Mbits}$$

$$\text{Burst to Mean Ratio} = \frac{5.12 \text{ Mbits}}{1.024 \text{ Mbits}} = 5.0$$

→ The initially chosen I²C won't be able to sustain this mean rate, as its peak exceeds the maximum capacity for I²C, So we can use SPI or AXI.

→ If we want to use I²C then we will need an "Asynchronous FIFO Buffer".

The buffer should hold either burst volume $= 5120 \text{ bits}$ or depth of 256 entries (20 bits/AER packet).

TASK 5

~~Implication AER is efficient choice when avg neural firing rate is below 50 Hz, as it exploits~~

Framebase $B = \frac{1024 \text{ bit}}{0.001 \text{ s}}$

$$= 1,024,000 \text{ bits/s}$$

$$= 1.024 \text{ Mbits/s}$$

AER to BW Ratio (50 Hz)

$$\text{Ratio} = \frac{1.024 \text{ Mbits}}{1.024 \text{ Mbits}}$$

$$\text{Ratio} = 1.0$$

Crossover Firing Rate.

$$B_{AER} = B_{frame} \Rightarrow N \times f_{cross} \times 20 = N \times 1000 \times 15 \text{ bit}$$

$$20 \times f_{cross} = 1000$$

$$\therefore f_{cross} = 50 \text{ Hz}$$